

AROMA

AI-powered Repository & Object Model Analyzer

Six AI-Powered Modules

Each powered by IBM Granite via watsonx.ai

IBM Bob Hackathon 2026 - Turn idea into impact faster
Built with IBM Bob IDE | Powered by IBM watsonx.ai Granite

1. Code Explorer - Architecture Mapping

The Question: What exists?

API Route: /api/analyze

Output: { files, dependencies, architecture, complexity, suggestions }

Key Features:

- AI identifies files, languages, line counts
- Function extraction with complexity scores
- Dependency mapping of import chains
- Architecture summary (MVC, microservices)
- 5 AI improvement suggestions

2. Flow Tracer - Interactive Code Flow

The Question: How does it work?
API Route: /api/chat
Output: { response: markdown }

Key Features:

- Conversational chat about codebase
- Flow tracing from endpoints to databases
- Multi-turn context (last 8 messages)
- Markdown technical responses
- Architecture explanations

3. Smart Refactor - Pattern Detection

The Question: What should I fix?

API Route: /api/refactor

Output: { suggestions: [...] }

Key Features:

- Deprecated patterns (var, callback hell)
- Code smell detection (magic numbers)
- Severity: Critical/High/Medium/Low
- Code diff with one-click copy
- Modern alternatives

4. Doc & Test Generator - Documentation Engine

The Question: How do I document?
API Route: /api/generate
Output: { title, doc, language }

Key Features:

- README.md with architecture + API ref
- API Documentation with schemas
- Vitest test suite with mocks
- JSDoc/TSDoc annotations
- 4 generation types

5. Health Dashboard - Quality Metrics

The Question: How healthy is my code?

API Route: (state-driven)

Output: 6 Recharts visualizations

Key Features:

- Health trend over 8 weeks
- Language distribution (Pie)
- Module complexity (Bar)
- Quality radar 6-axis
- AI-generated insights

6. Security Scanner - Vulnerability Detection

The Question: Is my code safe?
API Route: /api/security
Output: { findings: [...], score }

Key Features:

- OWASP Top 10 (10/10)
- SQL Injection (CWE-89), XSS (CWE-79)
- Hardcoded secrets (CWE-798)
- eval() injection (CWE-95)
- Score 0-100